

SMARTH SOOD

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Education

Indian Institute of Technology (BHU), Varanasi

Oct 2022 - May 2027

B.Tech + M.Tech (Integrated Dual Degree), Biomedical Engineering

CPI: 8.9/10

Mount Carmel School, ThakurDwara

2022

Class XII: Council for the Indian School Certificate Examinations (CISCE)

95.75%

Mount Carmel School, ThakurDwara

2020

Class X: Council for the Indian School Certificate Examinations (CISCE)

95.2%

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, Rust, SQL

Backend & Data: Node.js, Express, FastAPI, PostgreSQL, MongoDB, Redis, Apache Kafka, Apache Flink, Debezium, REST, WebSockets, JWT

Frontend: React, Next.js, Tailwind CSS, HTML/CSS

DevOps & Cloud: AWS (EC2, ECS, S3, Lambda, IAM), Docker, Kubernetes (k3s, k8s), GitOps, CI/CD, Prometheus, Grafana, Git, Linux

Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Object-Oriented Programming, Machine Learning

Experience

Wells Fargo | Software Engineer Intern

May 2026 – July 2026

- Architected a **fault-tolerant Java CDC pipeline** using **Debezium**, **Apache Kafka**, and **Apache Flink**, synchronizing **1M+ records** across distributed PostgreSQL databases with **sub-second p95 latency**.
- Built a **self-correcting LLM workflow** for automated XLSX table extraction using **LangChain** and **Apache POI**, leveraging **graph connectivity** to achieve **96% accuracy**.
- Deployed a production **document processing pipeline** with **Google Document AI** (OCR/Layout) which can process 100K+ financial docs with custom spatial reasoning, anomalies flagging achieving **97% accuracy** on **200+** docs.
- Reduced **80%** manual effort for analyzing **200+** page PDFs using **document chunking**, graph-based conflict resolution and numeric reconciliation.

Gokul Infocare Pvt. Ltd. | Python Developer

March 2024 – June 2024

- Engineered an **Image-to-MCQ pipeline** for automated document digitization and assessment generation, reducing processing time by **70%**.
- Developed an **agentic MCP** based **Hospital Appointment System** to streamline scheduling and appointment management, processing **100+ daily bookings**.
- Enhanced data collection using **Selenium-based web scrapers**, aggregating and cleaning **1,000+ records** for downstream analysis.

Projects

Drug Target Interaction – MLOps Platform

[Live Demo](#) | [GitHub](#)

- Engineered a no-code gene-expression ML platform with 6 models and 7-step preprocessing, using Redis/RQ, PostgreSQL, MLflow, and a 6-service Next.js/FastAPI multi-modal embedding architecture.
- Deployed multi-stage Docker builds on Kubernetes (k3s) with HPA on AWS Graviton; Designed JWT-secured, least-privilege IAM two-tier infrastructure to scale compute to zero and reduce costs by **~90%**.
- Built a GitOps CI/CD workflow with S3 artifacts and ConfigMaps; memory-profiled resource limits to stabilize the constrained multi-service cluster.

Tech Stack: Next.js, React, TypeScript, Node.js/Express, FastAPI, PostgreSQL, Redis, Docker, Kubernetes, AWS (EC2/S3/IAM), PyTorch, HuggingFace Transformers, MLflow, Embedding

Sentinel – Distributed Uptime Monitoring System

[Live Demo](#) | [GitHub](#)

- Architected a Rust/Tokio backend scheduling regional website checks via Redis sorted sets and Streams; bounded worker pool over 3 Redis connections with a 32-check semaphore, dispatching batches of up to 100 monitors per scheduler cycle, sustaining **around 9k checks/min at 200 ms p95** with 4vcpu and DB in Virginia.
- Persisted HTTP status, latency, size, DNS/TCP/TLS/TTFB timings, SSL validity, and failure diagnostics in PostgreSQL; exposed user-scoped JWT-authenticated APIs.

Tech Stack: Rust, Tokio, Poem, PostgreSQL, Diesel, Redis, Redis Streams, Redis Sorted Sets, JWT, Docker, Next.js, React, TypeScript

EEG-Based Biometric Authentication and Identification using Deep Learning — DRDO Project

[GitHub](#)

UG Research Project, IIT BHU – Supervised by [Dr. Jac Fredo Agastinose Ronickom](#).

- Crafted a CNN-based EEG biometric authentication system in TensorFlow, identifying 25 individuals across a two-run experiment spanning six cognitive domains, with an ICA, filtering, and segmentation preprocessing pipeline.
- Achieved 93.58% test accuracy and 94% F1-score using multi-task EEG features with session-wise generalization.

Tech Stack: Python, EEG Preprocessing, Deep Learning (CNN), NumPy, Streamlit, MNE-Python, Scikit-learn

Excalidraw-Style Real-Time Collaborative Drawing Platform

[Live Demo](#) | [GitHub](#)

- Full-stack collaborative drawing app supporting real-time multi-user sessions over WebSockets – infinite canvas, zoom/pan, live sync, and export – sustaining **50 ms sync latency across 1000+ concurrent users** per room.
- Secured REST and WebSocket endpoints with JWT authentication and custom authorization middleware, and managed a Turborepo + pnpm monorepo across 4 shared packages for unified types and database access, cutting CI build times **50%** via cached tasks.

Tech Stack: Next.js, WebSocket, PostgreSQL, Turborepo, pnpm, Canvas, AWS, Docker

Publications

Classification of ASD Improves with DFC of fMRI Compared to SFC

ISBS-RMBS 2024

- Conducted fMRI analysis on 112 subjects, distinguishing subjects with ASD from TD using sFC and dFC technique.
- Applied advanced methods like MLP, XGBoost and LR and achieved 88.76% accuracy on FC and 96.65% on dFC.

Identification of DEG in PDAC using Integrated Gradients and CNN

INCAM 2026

- Developed a CNN with dropout regularization for gene classification, achieving 84.6% accuracy and 84.4% F1-score; identified biomarkers (OR4K5, ZBTB7A) using K-Best selection and Integrated Gradients.

Honours & Achievements

Academic Achievements

- Granted Software Copyright for DTI platform; accepted for demonstration at the 36th EFMI MIE 2026 (Genova, Italy).
- Specialist on Codeforces (max rating 1471); Highest Rank 1111 in Codeforces Round 1034.
- Secured Platinum Rank in Flipr, 2nd Rank in BNY Mellon, and top-performer recognition in E6Data hackathon.

Extra-curricular and Leadership

- Joint Secretary of Aavaran (2024–2025) and member of DUGC and DPGC committees.
- Lead a contingent of 50+ people in Inter IIT Meet 2023,2024,2025 and hosted IIT BHU - MUN handling 200+ delegates across India.